

Stage 18: Notifications, Real-Time Dispatch & Support Ticket Helpdesk

1. Architectural Overview

The **Notifications, Real-Time Dispatch & Support Ticket Helpdesk Engine** delivers an omnichannel messaging, customer engagement, and issue resolution platform for the Alight International Multi-Vendor Marketplace. It powers real-time user notifications across multiple channels, system-wide broadcast alerts, customer support ticketing with order linkages, vendor assignment, and internal notes for support staff.



2. Core Entities & Database Structure

(`V44__notifications_and_support_ticket_enhancements.sql`)

2.1 Notifications (`notifications`)

- `user_id` : Target user recipient.
- `type` : `ORDER_CONFIRMED` , `ORDER_SHIPPED` , `ORDER_DELIVERED` , `RMA_UPDATE` , `QUOTE_UPDATE` , `REVIEW_ACTIVITY` , `TICKET_MESSAGE` , `SYSTEM_BROADCAST` , `PROMOTION` .
- `channel` : `IN_APP` , `EMAIL` , `SMS` , `PUSH` .
- `reference_id` & `action_url` : Deep-linking targets for UI interactions.
- `is_read` & `read_at` : Instant read/unread tracking.

2.2 Support Tickets (`support_tickets`)

- `ticket_number` : Formatted tracking reference (e.g., `TCK-2026-XXXXXX`).
- `category` : `ORDER_ISSUE` , `PAYMENT_FAILURE` , `RETURN_REFUND` , `PRODUCT_INQUIRY` , `VENDOR_DISPUTE` , `TECHNICAL_SUPPORT` .
- `priority` : `LOW` , `MEDIUM` , `HIGH` , `URGENT` .
- `status` : `OPEN` , `IN_PROGRESS` , `WAITING_ON_CUSTOMER` , `WAITING_ON_VENDOR` , `RESOLVED` , `CLOSED` .
- `vendor_id` & `order_id` : Multi-vendor entity linkage.
- `assigned_to` : Support agent assignment.
- `resolution_summary` & `resolved_at` : Formal resolution closure record.

2.3 Ticket Messages (`ticket_messages`)

- `sender_type` : `CUSTOMER` , `VENDOR` , `ADMIN` .
- `is_internal_note` : Flag for staff-only collaboration notes (omitted from customer-facing API views).
- `attachments` : JSONB array of photo/document URLs.

3. Communication & Security Architecture

1. Automatic Event Notifications:

- Creating a support ticket linked to a vendor automatically sends an in-app notification to the vendor's user account.
- Status updates or public message replies notify the counter-party immediately.

2. Internal Note Masking:

- `TicketResponse.fromEntity(ticket, includeInternalNotes)` strictly filters out `is_internal_note = true` when requested by a customer (`includeInternalNotes = false`), while revealing them to authorized admins and vendors (`includeInternalNotes = true`).

3. Multi-Tenant Ticket Isolation:

- Customers can only view, post to, and close tickets they opened.
- Vendors can only view and reply to tickets associated with their store.

4. Verification & Integration Suite

Verified via `NotificationsAndSupportIntegrationTest.java` :

- ✔ **Notification Center Lifecycle** : Notification creation, unread counting, recent summary, single mark-as-read, and mark-all-as-read.
- ✔ **Broadcast Dispatch**: System-wide announcements targeted to all or specific roles.
- ✔ **Support Ticket Creation & Auto-Notification**: Ticket initialization and vendor notification trigger.
- ✔ **Multi-Party Threading**: Customer replies, vendor responses, and staff internal notes with visibility isolation.
- ✔ **Ticket Lifecycle Transitions**: Status progression, agent assignment, and resolution summary logging.
- ✔ **Multi-Tenant Security**: Verification that customers cannot inspect or manipulate other users' tickets.
- ✔ **Ticket Statistics**: Summary aggregation across ticket statuses and priority levels.